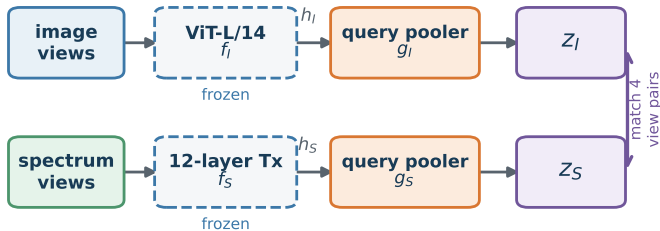


AstroJEPa learns a shared image--spectrum space while retaining modality-specific backbone states

A Pretrain → align

Frozen unimodal backbones; train 2.23M pooler parameters

unimodal SSL checkpoints



B Joint from scratch

Train both backbones and projectors end to end (392M)

random initialization

